

LVA-5 — Rotate Firebase CI token (printed to session transcript)

Status: Operator-blocked **Type:** Bug **Operator-Block-Details:**
WHAT: RuTracker CI Firebase token rotation WHY: Agent cannot rotate credentials; only the operator can run firebase logout / firebase login:ci UNBLOCK: Operator rotates the Firebase CI token (no token value ever committed, §11.4.10): [A] run "firebase logout" then "firebase login:ci" and write the new token to LAVA_FIREBASE_TOKEN in the gitignored .env (recommended); [B] mint a fresh token via "firebase login:ci --no-localhost" in a clean shell and update LAVA_FIREBASE_TOKEN in .env; [C] rotate via the Firebase console (Project Settings -> Service accounts) and update .env. Unblock signal: a Firebase distribute run authenticates with the new token and the previously-leaked token no longer authenticates. WHO: Operator **Severity:** P0

67th-cycle: the LAVA_FIREBASE_TOKEN default-expansion printed the token to the session transcript (NOT committed to git). Operator MUST rotate per §6.H clause 6 (firebase logout; firebase login:ci). **Source:** self-discovered — .lava-ci-evidence/sixth-law-incidents/2026-05-20-firebase-token-echo-leak.json

LVA-008 — C11 search_input NavBackStackEntry teardown crash (nested-NavHost lifecycle)

Status: Operator-blocked **Type:** Bug **Operator-Block-Details:**
WHAT: All 8 app-level candidate fixes are implemented + device-falsified (containerized-KVM gate, real JUnit XML evidence): inner-NavHost Activity-scoped LifecycleOwner, move-to-outer-NavHost, ON_STOP force-pop (NavTeardownGuard), nav-compose 2.9.1->2.9.8->2.10.0-alpha bump, LenientTeardownRule (uncatchable — process-fatal, not JUnit-catchable), atomic popUpTo replace, launchSingleTop dedupe, and full nested-NavHost-architecture collapse to single Activity-hosted multi-back-stack NavHost. Every attempt reproduces the identical IllegalStateException on C06+C11. Confirmed upstream androidx-navigation defect (b/244910446 family), no fixed version through 2.10.0-alpha04. WHY: Systematic-debugging Iron Law: 3+ failed fixes each revealing the same underlying defect in a different place = architectural problem, not grounds for a 9th ad-hoc app-level attempt. Minimal repro already authored and filing-ready at docs/lva008-upstream-repro/ (README.md, MinimalRepro.kt,

analysis.md). UNBLOCK: [A] AndroidX ships a navigation-compose release that fixes the nested-NavHost teardown ordering (b/244910446 family). [B] A Google/AndroidX engineer confirms a workaround this project hasn't tried. WHO: docs/lva008-upstream-repro/README.md (repro package ready to file); .lava-ci-evidence/sixth-law-incidents/2026-06-30-keystone-offmain-nav-and-lva008.json (latest incident record) **Severity: P1 Created-By: AI**

Challenge11ArchiveOrgAnonymousSearchTest crashes the app PROCESS at activity-destroy: IllegalStateException 'State must be at least CREATED to be moved to DESTROYED' on the inner search/search_input entry. Root-caused (2026-06-08): the inner nested NavController's host is the outer addNestedNavigation NavBackStackEntry, driven to DESTROYED out from under the inner controller while search_input is still INITIALIZED. FALSIFIED on device: nav 2.9.1->2.9.8, LenientTeardownRule (uncatchable process death), atomic popUpTo replace. Feature WORKS (result row renders pre-teardown). Candidate fixes ranked in incident JSON (inner NavHost Activity-scoped LifecycleOwner; move search to outer NavHost; ON_STOP pop). Forensics: .lava-ci-evidence/sixth-law-incidents/2026-06-08-navbackstackentry-teardown-crash-2.9.1-incomplete.json

LVA-120 — Pipeline 002 — firebase-distribute.sh --debug-and-release ships the RELEASE APK on DEBUG-channel evidence

Status: Operator-blocked **Type:** Bug **Operator-Block-Details:** WHAT: The combined --debug-and-release mode of scripts/firebase-distribute.sh uploads the release APK while evaluating the 6.AK/6.Z device-evidence gate against the debug channel, and skips the 6.AA staging gate entirely. Reproduced by tests/firebase/repro_mode_both_channel_gap.sh (exit 0 = gap still present). WHY: Changing the semantics of a LIVE distribution gate is an operator decision, not an autonomous one. Two mutually exclusive remedies exist and the choice determines what feature 002 task T043 is allowed to do. UNBLOCK: Operator picks one: [A] fix the channel resolution in scripts/firebase-distribute.sh so MODE=both evaluates the 6.AK/6.Z gate on the RELEASE channel and also runs the 6.AA staging gate; [B] retire --debug-and-release entirely and require the two-stage --debug-only then --release-only flow, which already evaluates both gates on the correct channels; [C] keep the flag but have T043 drive the two stages separately and never pass --debug-and-release. Unblock signal: tests/firebase/

repro_mode_both_channel_gap.sh exits 1 (gap GONE) and is converted into a standing regression test. WHO: Operator
Severity: P0 Created-By: AI

WHAT (verified verbatim in the real script this session, line by line, not inferred): scripts/firebase-distribute.sh line 56 - "--debug-and-release|--both) MODE="both"; shift". Line 204 - the constitution 6.AA staging gate is guarded on [["\$MODE" == "release" && -f "\$LAST_VERSION_DEBUG_FILE"]], so in combined mode it NEVER EVALUATES. Lines 322-323 - "case "\$MODE" in release) AK_CHANNEL="release" ;; *) AK_CHANNEL="debug" ;; esac", so the constitution 6.AK cycle-coverage gate (which also subsumes the 6.Z evidence presence, commit-SHA match and 24h freshness checks) is pointed at the DEBUG channel. Lines 435, 444 and 513 - the RELEASE APK is version-asserted, resolved and uploaded whenever MODE is release OR both. NET EFFECT: a combined distribute ships the R8-minified RELEASE APK while the only device-evidence gate that runs checks DEBUG-channel evidence. That is mechanically the same setup as the 1.2.19-1039 incident (the constitution 6.Z forensic anchor), in which a release APK crashed on every cold launch because release-variant behaviour was never gated. PRE-EXISTING: this is a property of scripts/firebase-distribute.sh and is NOT introduced by feature 002. REPRODUCTION: tests/firebase/repro_mode_both_channel_gap.sh. It NEVER executes firebase-distribute.sh (that would attempt a real Firebase upload); it greps the four real decision lines verbatim, evaluates that extracted logic per MODE, and exits 2 if those lines ever stop existing rather than silently testing a stale copy. Its exit semantics are inverted on purpose and documented in its header: 0 means the gap was REPRODUCED (expected today), 1 means it is GONE and the harness should be converted into a proper regression test, 2 means extraction failed and it can say nothing truthful - exit 2 is never read as "fixed". REPRODUCED OUTPUT: mode debug - release APK sent no, 6.AK channel debug, 6.AA staging gate no; mode release - release APK sent yes, 6.AK channel release, 6.AA staging gate yes; mode both - release APK sent YES, 6.AK channel DEBUG, 6.AA staging gate NO. The release row is the control that proves this is not a false alarm: --release-only DOES evaluate both gates on the correct channel. The gap is specific to the combined mode - precisely the mode an unattended pipeline would use. IMPACT ON FEATURE 002: task T043 was specified to invoke exactly --debug-and-release and would have inherited the hole; any amendment authorizing an unattended combined distribute would authorize exactly this. Source of truth: specs/002-build-test-distribute-pipeline/progress.yml and specs/002-build-test-distribute-pipeline/tasks.md.

LVA-121 — Pipeline 002 — HYPOTHESIS (not reproduced): phase-01-build.sh counts ARTIFACT lines, not distinct labels

Status: Queued **Type:** Task **Severity:** P3 **Created-By:** AI

UNCONFIRMED: this is a HYPOTHESIS, not a demonstrated defect. It was NOT reproduced during feature 002 and no failing case was constructed. Filed so it is not lost; it must be either reproduced (then re-filed as a Bug with RED evidence) or explicitly ruled out. CLAIM: scripts/pipeline/phase-01-build.sh counts ARTIFACT lines rather than distinct labels, so five duplicate labels would satisfy the ≥ 5 PASS condition. WHY IT WAS NOT PROMOTED TO A FINDING: not reachable through the real sub-scripts, each of which prints a given label at most once. To close, either construct a real producing path or record the argument that none exists. Source of truth: specs/002-build-test-distribute-pipeline/progress.yml and specs/002-build-test-distribute-pipeline/tasks.md.

LVA-122 — Pipeline 002 — HYPOTHESIS (not reproduced): recompute_evidence_summary would report zero for an unreadable dir

Status: Queued **Type:** Task **Severity:** P3 **Created-By:** AI

UNCONFIRMED: this is a HYPOTHESIS, not a demonstrated defect. It was NOT reproduced during feature 002 and no failing case was constructed. Filed so it is not lost; it must be either reproduced (then re-filed as a Bug with RED evidence) or explicitly ruled out. CLAIM: the `find ... 2>/dev/null` inside `recompute_evidence_summary` in `scripts/pipeline/lib/run-report.sh` would report zero Evidence Records for an unreadable phase directory, which `finalize_run_report` would then treat as zero rejections. WHY IT WAS NOT PROMOTED TO A FINDING: no realistic producing scenario was constructed. To close, either construct one or record why the phase directory cannot become unreadable mid-run. Source of truth: specs/002-build-test-distribute-pipeline/progress.yml and specs/002-build-test-distribute-pipeline/tasks.md.

LVA-123 — Pipeline 002 — HYPOTHESIS (not reproduced): phase-04 compose-file port parse is a no-op on an unquoted value

Status: Queued **Type:** Task **Severity:** P3 **Created-By:** AI

UNCONFIRMED: this is a HYPOTHESIS, not a demonstrated defect. It was NOT reproduced during feature 002 and no failing case was constructed. Filed so it is not lost; it must be either reproduced (then re-filed as a Bug with RED evidence) or explicitly ruled out. CLAIM: the compose-file port parse in scripts/pipeline/phase-04-live-verify-api.sh is a no-op when the LAVA_API_LISTEN value in docker-compose.yml is unquoted. WHY IT WAS NOT PROMOTED TO A FINDING: the real docker-compose.yml quotes the value, and a wrong port fails closed at curl rather than passing vacuously. To close, either construct a producing case or make the parse fail loudly on an unrecognized shape. Source of truth: specs/002-build-test-distribute-pipeline/progress.yml and specs/002-build-test-distribute-pipeline/tasks.md.

LVA-124 — Pipeline 002 — T040 human review gate — amend CLAUDE.md 6.AA + Seventh Law clause 3 for pipeline-run-report path

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: Constitutional amendment to root CLAUDE.md 6.AA and Seventh Law clause 3, drafted but NOT applied. Draft at specs/002-build-test-distribute-pipeline/constitutional-amendments-proposal.md. WHY: An agent must not amend the project constitution to relax an anti-bluff mandate it is itself subject to. This is a human review gate by construction. UNBLOCK: Operator picks one of the following CHOICES: [A] approve the drafted amendment for application as written, together with the matching .specify/memory/constitution.md MAJOR bump tracked separately as T041; [B] approve a NARROWER scope (for example: the alternate path applies only to the debug channel, leaving release distribution human-gated); [C] reject the amendment outright, in which case feature 002 task T043 stays permanently blocked and the pipeline never distributes. Unblock signal: the amendment is applied to CLAUDE.md in an operator-approved commit, or the operator records a decision not to amend. WHO: Operator **Severity:** P1 **Created-By:** AI

WHAT: feature 002 task T040 is a CONSTITUTIONAL AMENDMENT, not routine code review. It would amend root CLAUDE.md 6.AA (Two-Stage Distribute Mandate) and the Seventh Law clause 3 (Pre-Tag Real-Device Attestation) per research.md R-001, adding an alternate-path clause under which a Pipeline Run Report for the exact commit SHA and versionCode substitutes for human debug-then-release confirmation, scoped narrowly to distributions this pipeline itself produces. It must include a forensic anchor, a migration plan and a version-bump rationale per the constitution own Amendment procedure. BLAST RADIUS: this is a RELAXATION of an existing anti-bluff principle whose forensic anchor is the 1.2.19-1039 cold-start-crash incident. It also interacts directly with the open firebase-distribute combined-mode finding filed alongside this item: authorizing an unattended combined distribute would authorize exactly the debug-evidence-gates-a-release-APK gap. Review-ready DRAFT text for this amendment is at specs/002-build-test-distribute-pipeline/constitutional-amendments-proposal.md (with .html and .pdf siblings). The draft is NOT applied: CLAUDE.md and .specify/memory/constitution.md are byte-identical to their pre-feature baselines. BLOCKS: feature 002 phase 5 (US3 distribute), specifically task T043. Source of truth: specs/002-build-test-distribute-pipeline/progress.yml and specs/002-build-test-distribute-pipeline/tasks.md.

LVA-125 — Pipeline 002 — T041 human review gate — .specify/memory/constitution.md MAJOR bump 1.1.0 to 2.0.0 matching T040

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: Paired constitution-file bump for the T040 amendment, drafted but NOT applied. .specify/memory/constitution.md is byte-identical to its pre-feature baseline. WHY: Same human review gate as T040; the two are one governance change. UNBLOCK: Operator picks one of the following CHOICES: [A] approve T040 and land this MAJOR bump in the SAME reviewed commit (the recommended pairing); [B] approve T040 but land this bump as an immediate follow-up commit; [C] reject T040, in which case this bump is abandoned and the file stays at 1.1.0. Unblock signal: .specify/memory/constitution.md reads 2.0.0 with an updated Sync Impact Report, or the operator records the rejection. WHO: Operator **Severity:** P1 **Created-By:** AI

WHAT: feature 002 task T041 updates .specify/memory/constitution.md to match the T040 amendment with a MAJOR version bump 1.1.0 to 2.0.0, per that file own Versioning policy - MAJOR because this is a RELAXATION of an existing principle, not an addition - and updates the Sync Impact Report header

accordingly. Paired with T040: neither lands alone. Review-ready DRAFT text for this amendment is at specs/002-build-test-distribute-pipeline/constitutional-amendments-proposal.md (with .html and .pdf siblings). The draft is NOT applied: CLAUDE.md and .specify/memory/constitution.md are byte-identical to their pre-feature baselines. BLOCKS: feature 002 phase 5 (US3 distribute), specifically task T043. Source of truth: specs/002-build-test-distribute-pipeline/progress.yml and specs/002-build-test-distribute-pipeline/tasks.md.

LVA-126 — Pipeline 002 — T048 human review gate — Decoupled Reusable Architecture carve-out for automatic pin advancement

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: Constitutional amendment to the Decoupled Reusable Architecture rule in root CLAUDE.md, drafted but NOT applied. WHY: Relaxing the never-automatic submodule-pin rule affects 16-plus upstreams and the project own governance submodule. Human review gate by construction. UNBLOCK: Operator picks one of the following CHOICES: [A] approve the drafted carve-out as written; [B] approve a NARROWER carve-out (for example: automatic advancement permitted only for an explicit operator-supplied allow-list of submodules, never a default-all); [C] reject the carve-out, in which case feature 002 task T055 stays permanently blocked and submodule advancement remains a manual operator action. Unblock signal: the carve-out is applied to CLAUDE.md in an operator-approved commit, or the operator records a decision not to amend. WHO: Operator **Severity:** P1 **Created-By:** AI

WHAT: feature 002 task T048 is a CONSTITUTIONAL AMENDMENT to the Decoupled Reusable Architecture rule in root CLAUDE.md, carving out this pipeline automated submodule-pin-advance step as an explicit, documented exception to "Submodule fetch/pull is an EXPLICIT operator action, never automatic". BLAST RADIUS: 16-plus submodule upstreams with automatic pin advancement. The wave-5 defect filed alongside this item is the concrete demonstration of why this amendment needs careful scoping: the implementing script defaulted to every submodule and would have auto-advanced the constitution governance submodule itself until a default-deny list was added. Review-ready DRAFT text for this amendment is at specs/002-build-test-distribute-pipeline/constitutional-amendments-proposal.md (with .html and .pdf siblings). The draft is NOT applied: CLAUDE.md and .specify/memory/constitution.md are byte-identical to their pre-feature baselines. BLOCKS: feature 002

phase 6 (US4 repository closure), specifically task T055. Source of truth: specs/002-build-test-distribute-pipeline/progress.yml and specs/002-build-test-distribute-pipeline/tasks.md.

LVA-127 — Pipeline 002 — T049 human review gate — .specify/memory/constitution.md update matching T048

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: Paired constitution-file update for the T048 carve-out, drafted but NOT applied. WHY: Same human review gate as T048; the two are one governance change. UNBLOCK: Operator picks one of the following CHOICES: [A] approve T048 and land this update in the SAME reviewed commit as T041 under one MAJOR bump; [B] approve T048 and land this update as a separate follow-up MAJOR bump; [C] reject T048, in which case this update is abandoned. Unblock signal: .specify/memory/constitution.md records the carve-out with a bumped version and an updated Sync Impact Report, or the operator records the rejection. WHO: Operator **Severity:** P1 **Created-By:** AI

WHAT: feature 002 task T049 updates .specify/memory/constitution.md to match the T048 carve-out - the same MAJOR version bump as T041 if landed together, or a follow-up MAJOR if landed separately. Review-ready DRAFT text for this amendment is at specs/002-build-test-distribute-pipeline/constitutional-amendments-proposal.md (with .html and .pdf siblings). The draft is NOT applied: CLAUDE.md and .specify/memory/constitution.md are byte-identical to their pre-feature baselines. BLOCKS: feature 002 phase 6 (US4 repository closure), specifically task T055. Source of truth: specs/002-build-test-distribute-pipeline/progress.yml and specs/002-build-test-distribute-pipeline/tasks.md.

LVA-128 — Pipeline 002 — T054 human review gate — mandatory review of advance-all-submodules.sh before ANY real invocation

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: scripts/advance-all-submodules.sh has never run against a real submodule or upstream and must not until this review completes. Fixture-only TDD evidence: tests/pipeline/test_advance_all_submodules.sh, 36/36. WHY: Highest blast radius in feature 002 - automatic pin advancement across 16-plus real upstreams. plan.md Human Checkpoint #2 makes the review

mandatory and prior to first real invocation. UNBLOCK: Operator picks one of the following CHOICES: [A] a human reviews scripts/advance-all-submodules.sh against the criteria above, records approval, and authorizes a first run against a DISPOSABLE clone per task T056; [B] a human reviews it and requires specific changes, which are made and re-reviewed before any real invocation; [C] the operator declines to authorize automated submodule advancement at all, in which case feature 002 task T055 is descoped and the script is never invoked against a real upstream. Unblock signal: the review verdict is recorded, and for CHOICE [A] the disposable-clone run of task T056 completes. WHO: Operator or a designated human reviewer **Severity: P0**
Created-By: AI

WHAT: feature 002 task T054 is a MANDATORY code review of scripts/advance-all-submodules.sh before any invocation against a real submodule upstream, per plan.md Human Checkpoint #2. This is the highest-blast-radius new component in the feature: 16-plus submodule upstreams with automatic pin advancement, plus commit-and-push of each submodule own local modifications. CURRENT STATE, stated honestly: the T050-T053 TDD cycle against DISPOSABLE FIXTURE repos is COMPLETE - 36/36 assertions green, independently re-run by the orchestrating session - and real submodules were confirmed untouched (git status --porcelain over submodules/ and constitution/ empty, git diff --cached empty). scripts/advance-all-submodules.sh has NEVER been run against a real submodule or a real upstream. REVIEW MUST COVER at minimum: the GOVERNANCE_DENY default-deny list added by the wave-5 fix and the fact it is deliberately not overridable via LAVA_ADVANCE_SUBMODULES; the 7-step ordering from research.md R-005 (fetch, compare, no-op or checkout-advance, rebuild-and-test, discard-and-report on failure, commit/push the submodule own local modifications, update the parent pin); and the refusal paths REJECTED_BREAKING_CHANGE, REJECTED_PUSH_CONFLICT and REFUSED_GOVERNANCE_DENY. Source of truth: specs/002-build-test-distribute-pipeline/progress.yml and specs/002-build-test-distribute-pipeline/tasks.md.

LVA-129 — Pipeline 002 — T062 human review gate — full end-to-end pipeline run on a disposable branch before trusting it

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: The full five-phase end-to-end orchestrator run has never been executed. It must first run on a disposable branch, never on real master. WHY: plan.md Human Checkpoint #3. A first full run touches Gradle, systemd and an emulator on the real host and

must be supervised on a throwaway branch. UNBLOCK: Operator picks one of the following CHOICES: [A] a human runs quickstart.md Scenario 2 end to end on a DISPOSABLE branch with all five wired phases and reviews the resulting report.json; [B] a human runs it phase by phase on a disposable branch, reviewing each phase result before proceeding to the next; [C] the operator declines the end-to-end run, in which case scripts/pipeline-build-test-distribute.sh remains untrusted and must not be pointed at real master. Unblock signal: a run report for the disposable branch validates against contracts/pipeline-run-report.schema.json and is recorded, or the operator records the decision to decline. WHO: Operator or a designated human reviewer **Severity: P1 Created-By: AI**

WHAT: feature 002 task T062 requires a full end-to-end run of quickstart.md Scenario 2 (all phases, no shortcuts) on a DISPOSABLE branch, per plan.md Human Checkpoint #3, before scripts/pipeline-build-test-distribute.sh is trusted to run against real master. CURRENT STATE, stated honestly: the full end-to-end orchestrator run across all five wired phases has NOT been performed. It invokes Gradle, systemd and an Android emulator, and was deliberately not run because it is this review gate and because it would have contended with parallel agents already using those toolchains. It is not claimed as done anywhere. SCOPE NOTE: only phases precondition, build, test, install_boot and live_verify are wired; distribute, docs and closure are deliberately not wired pending T040/T041 and T048/T049, and asking for them via --until is a usage error rather than a silent no-op. Source of truth: specs/002-build-test-distribute-pipeline/progress.yml and specs/002-build-test-distribute-pipeline/tasks.md.